
TECHNICAL SKILLS

Core : C++, Python, ROS 2 (roscpp/rospy), TF/URDF, DDS/RTPS, Linux, Git, Docker, CI / CD

Perception & Estimation : OpenCV, PyTorch, Kalman (EKF/UKF), YOLO, TensorRT, PCL/Open3D, NumPy, Pandas

Navigation & Control : Nav2, OMPL, MPPI/MPC, PID Cascades, Kinematics-based Control, Precision Docking

Sensors & Calibration : LiDAR 2D/3D, RGB-D, IMU, Odometry (Wheel, LiDAR-inertial, Visual), Sensor Fusion

Simulation & Motion : Gazebo/IGN, RViz, MuJoCo, CARLA, Moveit2

PROFESSIONAL EXPERIENCE

Software & Automation Developer

Rashmi Enterprises | Jan 2021 - July 2023

India

A specialized firm providing industrial printing and automated packaging solutions.

Technologies: ROS 2, C++, Python, Gazebo, OpenCV, PCL, Git, CI/CD

- **Accelerated deployments ~30%** by architecting a modular ROS 2 stack (perception - fusion/planning - control) to replace a monolith; improved reuse and testability.
- **Delivered 95% accuracy @ 20 FPS** real-time anomaly detection (OpenCV) with deterministic launches and stable TF frames.
- **Unblocked a system bottleneck** by developing & stress-testing **multi-robot navigation for a 5-AMR fleet** in Gazebo, automating material flow.
- **Hardened production reliability** with reproducible Linux/Docker builds and runbooks; **throughput increased 15%, operator interventions decreased 40%**.

PROJECTS

ADAS Stack – Lane Keeping & Emergency Braking

[GitHub](#) | [Demo](#)

ROS2, Python, Carla

- Lane Keeping Control - Maintained **≤ 0.22 m lateral error** at **60 km/h** with pure-pursuit, adaptive look-ahead, and PID corrections; **p95 steering-command latency 12 ms**.
- Multi-sensor Fusion - Fused LiDAR & RGB (front), and dual-radar (L/R) & RGB (rear) with an EKF plus Mahalanobis/IoU gating; **reduced camera-only false positives 36%** in dusk/rain and **ID-switches 28%** across **40+ scenarios**.
- AEB - TTC-gated AEB achieved **95% collision-avoidance** over 40 dynamic scenarios; **p95 perception latency 42 ms** and **end-to-stop response 430 ms**.

ATLAS - Autonomous Warehouse Picking and Inventory Management

[GitHub](#) | [Demo](#)

ROS2, C++, Python, PCL, Gazebo

- Mapping & State Estimation - Fused IMU, wheel odom, and VO/LIO for stable map/odom/base_link; met **ATE < 0.15 m over 100 m** and **loop-closure drift $< 1\%$** .
- Perception & Tracking - **PCL** pipeline (pass-through, 1 cm voxel, SOR; RANSAC plane; Euclidean clustering) hit **$\geq 85\%$ correct segmentation** at **< 40 ms @ 30 Hz**; per-object KF kept **ID-switch $< 5\%$** on 2–3 min patrols.
- Planning & Task Execution - Nav2 with OMPL BIT* global + MPPI local achieved **plan P50 < 150 ms (P95 < 500 ms)** and **RMS tracking error < 0.10 m @ ~ 1 m/s**; BehaviorTree.CPP reached **$\geq 85\%$ end-to-end patrol/pick success**.

HELIOS - SAR Quadcopter Control

[GitHub](#) | [Demo](#)

ROS2, C++, Gazebo

- Real-time Controller Design - Implemented **200 Hz cascaded PID (position - velocity - attitude - rate)** in C++ commanding thrust + body torques, with anti-windup and clamping; **no heap allocations post-init**; **p95 loop jitter 2.1 ms** on **PREEMPT_RT Linux**.
- Robustness and Performance - Validated failsafes across 20 injected faults (e.g., sensor dropouts), achieving **controlled descent within 0.6 m** touchdown radius while keeping **attitude error $< 2.5^\circ$** ; nominal hover **RMS 0.22 m**.

Real Steel – Vision-to-Motion Robot Tracker

[GitHub](#) | [Demo](#)

ROS2, Python, MuJoCo

- Low-Latency Motion Retargeting - Engineered a vision-to-control pipeline that maps human keypoints (MediaPipe) to a **7-DOF robot arm**, achieving an **85ms median end-to-end latency (p95 110ms)**.
- Smooth & Robust Motion - **Smoothed motion by $\sim 90\%$** using a **joint-limit-aware IK solver** and integrated the full pipeline for responsive sparring in simulation.

EDUCATION

Iowa State University

Ames, IA

MS, Computer Science (GPA: 3.72/4)

Aug 2023 - Aug 2025